Appendix A

Silicon Ingot and Wafer Manufacturing*

POLYCRYSTALLINE ("POLY") SILICON PRODUCTION

In a two-stage process, quartzite (silica sand) is reduced in the presence of coke in an electric arc furnace. Molten metallurgical-grade silicon (Si) is drawn from the furnace and blown with oxygen or an oxygen-chloride mixture to reduce the levels of aluminum, calcium, and manganese impurities (approximately 99% pure silicon). The next step involves reacting silicon with hydrogen chloride to form trichlorosilane ($SiHCl_3$) in the presence of a copper-containing catalyst. Trichlorosilane is then decomposed in a hydrogen atmosphere with electric current (1100–1160°C) to form pure free silicon ingots from the starter silicon seed crystals.

These polycrystalline ingots are the basic material melted in a crucible to feed the growth of single crystal ingots which are the basis of silicon device manufacturing.

SINGLE CRYSTAL INGOT GROWTH

Czochralski

Czochralski (Cz) and float zone technologies dominate the production techniques used to manufacture single crystal silicon for the semiconductor industry. Cz, the most widely used crystal growing method, begins by placing semiconductor grade polycrystalline silicon in a fused silicon quartz

* This information is compiled primarily from references F1, F2, and F3 shown in Appendix F.

crucible which rests in a carbon susceptor (Fig. A1). Heating this array to 1200°C with either radio-frequency (RF) or resistance heating in an inert argon atmosphere melts the silicon and keeps the material just above the melting point. A seed crystal, or single crystal, of the desired crystallographic orientation is dipped into the melt to form the crystal. Rotating the seed and the crucible in opposite directions, and at the same time withdrawing the seed, permits the crystal to grow. Careful control of temperature, rotation speed, and vertical withdrawal determines ingot size, and therefore, possible diameter. The most common wafer diameters are: 4" (100 mm), 5" (125 mm), and 6" (150 mm); and the latest integrated circuit manufacturers are using 8" (200 mm). It is currently unclear whether the progression will go to 10" (250 mm) or directly to 12" (300 mm) wafers as the next generation of wafer diameter.

Depending on the type of crystal being grown, different atmospheres and pressures are maintained in the growth chamber. Inert, oxidizing or reducing atmospheres from vacuum (10^{-5} torr) to high pressure (300 psi) are used, with vacuum atmosphere growth method predominating.

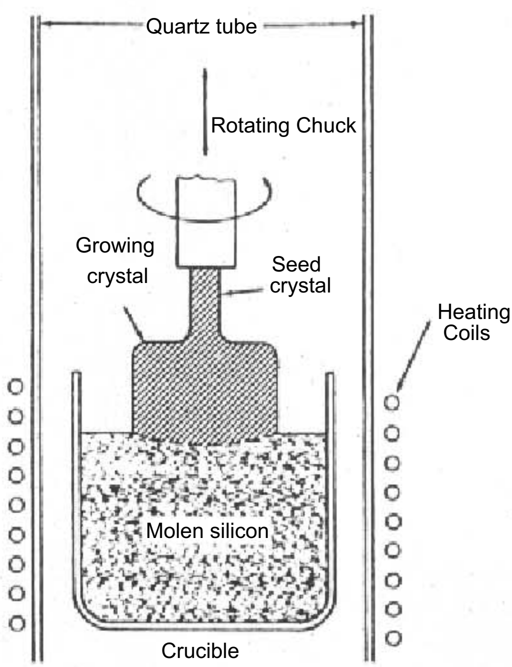

Figure A1. Czochralski (Cz) ingot growth apparatus. *(© 1981, CAL-OSHA, used with permission.)*

Float Zone

With the float zone technique (Fig. A2), a rod of polycrystalline silicon 50–100 cm long is attached to a chuck at the top of an inert-atmosphere vacuum chamber and a seed crystal is attached to a shaft at the chamber's bottom. Applying current to an RF heating coil wound closely around the base of the rod forms a molten zone about 2 cm long. Fusing the resulting droplet with the seed brought up from below begins crystal formation. Moving this molten zone, either by lowering the rod through the stationary coil or passing the coil up the stationary rod, purifies the material and establishes the crystal structure. Melted silicon is kept in place by surface tension, which limits the diameter of the crystal and the height of the float zone. Impurities are either retained in the molten zone or evaporated into the vacuum and removed.

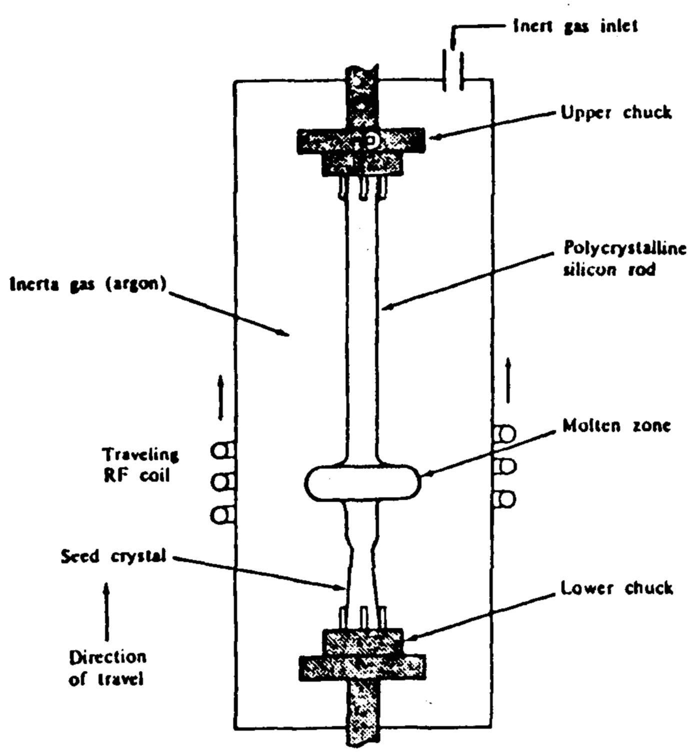

Figure A2. Float zone (FZ) ingot growth apparatus. *(© 1981, CAL-OSHA, used with permission.)*

Adding controlled amounts of impurities during the growth process creates the necessary electrical properties within the silicon's crystal structure. Impurities such as boron are added to establish a p-type resistivity; antimony, phosphorus, and arsenic are added to produce an n-type. In the Czochralski process, dopants are usually added to the molten silicon in the form of pure elements or concentrated silicon alloys. With the float zone method, diluted concentrations of phosphine or diborane gas are added during the pass on the rod or when the poly rod is manufactured.

INGOT EVALUATION

Ingot Cropping

The ends or tails of the single-crystal ingot are removed using a water-lubricated single-bladed diamond saw, with various coolants added to the water.

Ingot Grinding

The ingots are then ground to a uniform diameter over their total length by a water-lubricated grinder, again with various coolants added to the water.

Ingot Chamfering

The ends of the cropped and ground ingots are chamfered (beveled), using a dry belt sander, which reduces the possibility of shattering the ingot.

Ingot Flatting

The orientation of the crystalline structure within the silicon ingot is important for subsequent processing. This is indicated by wet grinding a flat surface on the exterior of the ingot, using various coolants added to the water. The crystalline structure orientation is determined by x-ray diffraction analysis.

Seed Slicing

The single-crystal silicon seed crystals, from which the crystalline structure of the ingot is obtained, are wet-sliced for uniformity using a mechanical saw and water to which various coolants have been added.

X-ray Diffraction

An x-ray diffraction unit is used to focus x-ray beams into the silicon ingots for determining its crystalline structure. A photograph is taken of the crystal orientation and is used for identifying the surface to be ground in ingot flatting.

Etching

Both the ingots and seed crystals are etched with wet acid mixtures to help delineate crystalline structure and for surface cleaning. The major categories of etchants for silicon ingots and seed crystals are:

- Sirtl etch: hydrofluoric and chromic acids ($HF + CrO_3$)
- Dash etch: hydrofluoric, nitric, and acetic acids (HF, HNO_3, CH_3COOH)
- Typical ratios: 2:5:1
3:8:2
1:5:1
- Secco etch: hydrofluoric acid and potassium dichromate

  ($HF + K_2CR_2O_7$)

  NOTE: Standard concentrations: HF = 49%; HNO_3 = 67%; CH_3COOH (acetic acid) = 36.5–99%

SLICING

Ingot Mounting

The ingots are mounted to a graphite carrier rod by the use of a two-component epoxy resin system. This attachment facilitates the mounting of the ingots to the wafer slicing saw.

Wafer Slicing

Ingots are sawed into individual wafers through the use of multiple inner diameter (ID) diamond blade saws. This operation is done wet with the use of lubricants and, in some processes, the wafers are stored in plastic reservoirs containing water or methanol. The thickness of the wafers after slicing depends on the initial diameter of the ingot and intended usage. Approximately 28 wafers are cut from each inch of the ingot.

Wafer Washing

Wafers are then transferred from the plastic reservoirs containing methanol into wash tanks with a proprietary soap mixture, ~10% sodium hydroxide (NaOH) mixture, and finally water rinsed.

Wafer Lapping

Wafers are lapped on machines exerting a set rotational speed and pressure. A lapping solution is fed onto the lapping surface and constitutes a slurry of colloidal silica, water, and proprietary wetting agents, which sometimes contain ethylene glycol, morpholine, and hexavalent chromium. Lapping is intended to remove the rough outer surface of the wafer caused by sawing.

Edge Rounding

The edges of the individual wafers are rounded by the use of wet automatic grinders.

Wafer Etching

Wafers are placed in fluorinated plastic cassette holders and loaded into either manual chemical etch tanks or into an automated etching machine. The etching solutions contain nitric acid, hydrofluoric acid, and acetic acid in the following ratios:

HNO_3	HF	and	CH_3COOH
5	2		1
8	2		3

The wafer etching process removes external surface damage to the wafers, and reduces their thickness.

POLISHING

Wafer Polishing

The wafers are initially mounted onto a metal carrier plate. The carrier is then attached by vacuum to the polishing machine. Generally, a two-step process is performed—first, *stock removal*, which provides an initial polish and decreases wafer thickness, and secondly, a *finish* polish to achieve a mirror-like exterior surface. A polishing solution containing sodium hydroxide, colloidal silica, and water is automatically dispensed during polishing.

Carrier Stripping

If carrier pads must be stripped from the metal carrier plates, chemical stripping is done with solvents, such as methylene chloride, methyl ethyl ketone (MEK), or a glycol ether mixture.

Polish Cleaning

This cleaning removes any particles or residue that may be on the exterior surface of the polished wafer. Separate baths of chemical solutions are used, and may contain:

1. Twenty-five percent hydrofluoric acid (HF)
2. Combination of soaps and/or oxidizers, such as dilute hydrogen peroxide (H_2O_2)
3. Cleansing agents

FINAL PROCESSING

Final Cleaning

The last operation for cleaning the wafer surface involves:

1. Scrubbing with ammonia (NH_3) and water
2. Microscopic inspection
3. Visual inspection

Epitaxial (Optional)

An optional step is to deposit a thin single-crystal silicon film containing a dopant material—an epitaxial layer. This layer has different electrical properties than the base silicon wafer.

Oxidation (Optional)

Another optional step is the formation of an oxide layer on the wafer. The silicon dioxide layer helps protect the wafer from scratches and contamination while it is being transported.

Tray Cleaning

In some operations, plastic trays which are to be used for the final packaging of the finished wafers are first cleaned in a vapor degreaser using a solvent, and then bagged.

Final Packaging

The cleaned, tested and approved wafers are loaded into the plastic trays and cellophane shrink-wrapped for contamination control and shipment.